

KATANA IMRAN TARIQ

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PERSONAL PROFILE

Second-year BSc (Hons) Computer Science student at Aston University, predicted to graduate with a First Class degree. Experienced in building data pipelines, quantitative models, and scalable backend systems applied to real financial problems. Developed a commodity risk analytics platform processing 60+ years of market data integrating time-series forecasting, FX risk modelling, revenue simulation, and ERA5 satellite climate data across a Python and FastAPI stack. Interested in how software engineering at scale powers financial markets infrastructure, and in building systems that translate complex data into actionable insight.

KEY COMPETENCIES

Java, Python, SQL, PHP, JavaScript, HTML, CSS, React 18, Vite, Tailwind CSS, Laravel, Tkinter, Swing, Matplotlib, GitHub, MySQL, SQLite, PostgreSQL, VS Code, IntelliJ, Prophet, Pandas, NumPy, Scikit-learn, FastAPI, Streamlit, REST APIs, xarray, Plotly, sql.js (WebAssembly), IndexedDB, ExchangeRate-API

EDUCATION AND QUALIFICATIONS

BSc (Hons) Computer Science · Year 2 · Predicted: First Class

Aston University, Birmingham, UK | Sept 2024 – Present

- Progressed directly into Year 2 having completed the International Year One pathway (ONCAMPUS Aston) with a First Class grade — skipping the standard first year entry route
- Year 2 modules: Data Structures & Algorithms, Artificial Intelligence, Systems Analysis, Information Security, Software Engineering, Programming Language Concepts, Object-Oriented Design, Computational Mathematics
- Year 1 foundations: Java programming, full-stack web development (HTML, CSS, JavaScript, PHP, SQL), GitHub version control, and four major portfolios covering CRUD operations, UI/UX design, backend workflows, and database integration

A-Levels (Cambridge Assessment International Education)

Kabojja International School, Uganda | Jun 2024

Mathematics: B · Business Studies: B · Physics: C

IGCSE (Cambridge Assessment International Education)

Kabojja International School, Uganda | Jun 2022

Physics: A · Mathematics: A · English Language: A · Business Studies: B · Chemistry: B · ICT: B · French: B · Geography: B

WORK EXPERIENCE

JPMorgan Chase & Co. (via Forage) — Quantitative Research Job Simulation

Feb 2026 | Remote

- Investigated and analysed large datasets to identify trends and patterns, developing data processing skills applicable to business intelligence
- Applied structured analytical frameworks to model real-world scenarios, translating raw data into actionable technical outputs
- Gained exposure to industry-standard quantitative methods, strengthening problem-solving and systems thinking skills

Student Representative

ONCAMPUS Aston | Sept 2024 – Jun 2025

- Strengthened communication channels between students and academic staff by collecting and presenting structured feedback
- Led initiatives that transformed passive lecture formats into interactive sessions, advocating for Kahoot-based assessments that improved engagement across the cohort
- Proposed and secured a QR code onboarding system for new international students, resulting in a thriving weekly student football group
- Developed leadership, public-speaking, and stakeholder communication skills through regular formal presentations to faculty

RELEVANT PROJECTS

Uganda Coffee Price Forecasting & Commodity Risk Platform | github.com/KatanaTariq/uganda-coffee-platform | Jan 2026 – Present

Python · Pandas · Prophet · FastAPI · Streamlit · Scikit-learn · ERA5 · Plotly · SQLite

- Built a multi-source quantitative analytics platform processing 796 months of Arabica and Robusta commodity price data from the World Bank, Bank of Uganda FX rates, and ERA5 satellite climate data merging three independent datasets into a unified analytical pipeline
- Engineered a time-series price forecast model using Prophet with MAE, RMSE, and MAPE evaluation metrics; implemented 12-month forward projections with 95% confidence intervals to quantify forecast uncertainty
- Developed an FX risk and revenue simulation engine modelling how simultaneous shocks to coffee price, export volume, and UGX/USD exchange rate affect exporter revenue including a five-scenario stress test framework mirroring industry-standard risk management methodology
- Integrated 256 months of ERA5 satellite rainfall, temperature, and soil moisture data as leading indicators of supply shocks; computed a drought detection flag and crop health score to produce a supply risk signal preceding price movements by 6–9 months

FeeTracker — Family Financial Management System | github.com/KatanaTariq/school-fees-manager | 2025

React 18 · Vite · Tailwind CSS · SQLite (sql.js / WebAssembly) · IndexedDB · JavaScript (ES2022) · REST API

- Designed and built a full-stack web application to manage school fee payments and rent for a household of 6 dependents across multiple currencies (USD, GBP, UGX), with real-time foreign exchange conversion via ExchangeRate-API
- Architected a normalised relational database schema (8 tables, foreign key constraints, cascading deletes) running entirely in the browser via WebAssembly (sql.js) with IndexedDB persistence — no server required
- Implemented multi-currency financial ledger with live FX rate fetching, historical rate preservation, and automated installment amortisation engine splitting tuition into weighted payment tranches
- Engineered overdue aging logic and deadline reminder system with day-precision alerting, plus a mobile-responsive dashboard with real-time state management using React hooks

CERTIFICATES

- JPMorgan Chase & Co. Quantitative Research Job Simulation – Forage (Feb 2026)

SKILLS AND INTERESTS

Languages: English (fluent)

Societies: Aston Computer Science Society, Aston Eagles Basketball Club

Interests: Technology for social impact — particularly how digital infrastructure and connectivity can drive economic development and agricultural innovation in emerging markets. Fitness and basketball.